Monte Carlo Methods

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January 2026

Integration via Monte Carlo Methods

On the topic of solving integrals (last week solved by using recursions). Consider

$$\int_0^1 x^3(sin(x^4) + cos(x^5))e^{-x^6}dx$$

$$\int_0^\infty cos(x^7)e^{-5x}dx$$

$$\int_{-\infty}^\infty cos(x^7)e^{-x^2/2}dx$$

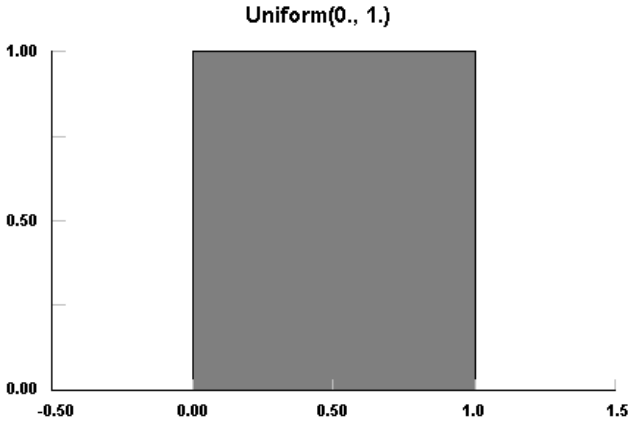
Can you solve these with the methods from differential calculus (integration by parts, substitution)? Probably not.

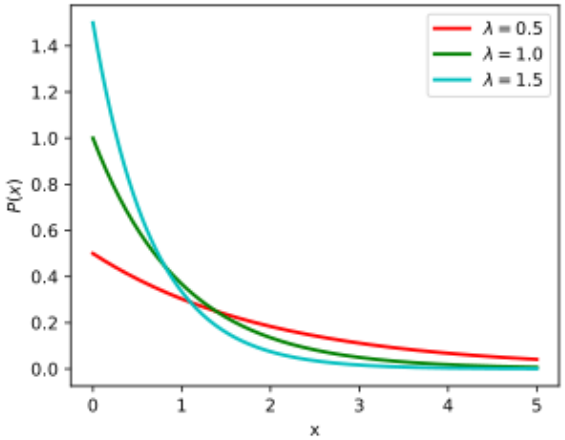
More Concepts in Probability

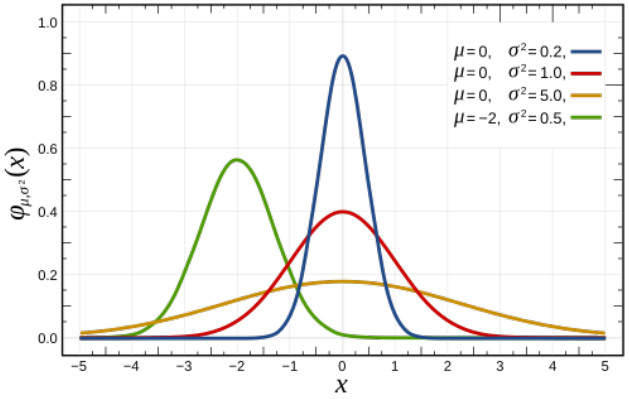
Uniform [L,R] $X \sim \mathrm{Uniform}(L, R), \quad f_X(x) = \begin{cases} \dfrac{1}{R-L}, & L \le x \le R \\ 0, & \text{otherwise} \end{cases}$

Exponential(λ) $X \sim \mathrm{Exponential}(\lambda), \quad f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \ge 0 \\ 0, & \text{otherwise} \end{cases}$

Normal (N(μ,σ^2)) $X \sim N(\mu, \sigma^2), \quad f_X(x) = \dfrac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\dfrac{(x-\mu)^2}{2\sigma^2}\right)$







Even More Concepts in Probability

Theorem (Weak Law of Large Numbers)

Let $X_1, X_2, \ldots$ *be i.i.d. random variables with* $\mathbb{E}[X_1] = \mu$ *and* $\mathrm{Var}(X_1) = \sigma^2 < \infty$. *Define the sample mean*

$$\overline{X}_n = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

Then, for every $\varepsilon > 0$,

$$\lim_{n \to \infty} \mathbb{P}\left(\left|\overline{X}_n - \mu\right| > \varepsilon\right) = 0.$$

Equivalently, $\overline{X}_n \xrightarrow{P} \mu$ *as* $n \to \infty$.

Intuitively: "Means of samples converge to the actual mean for large n"

Applying Monte Carlo Methods

Define,

$$X_i = U_i^3(sin(U_i^4) + cos(U_i^5))e^{-U_i^6}, \qquad U_i \sim Uniform[0,1]$$

$$Y_i = cos(V_i^7)e^{-5V_i}, \qquad V_i \sim Exponential(5)$$

$$Z_i = cos(W_i^7)e^{-W_i^2/2}, \qquad W_i \sim Normal(0,1)$$

So by the definition of expected value,

$$E[X_i] = \int_0^1 x^3(sin(x^4) + cos(x^5))e^{-x^6}dx = \mu_{X_i}$$

$$E[Y_i] = \int_0^\infty cos(x^7)e^{-5x}dx = \mu_{Y_i}$$

$$E[Z_i] = \int_{-\infty}^\infty cos(x^7)e^{-x^2/2}dx = \mu_{Z_i}$$

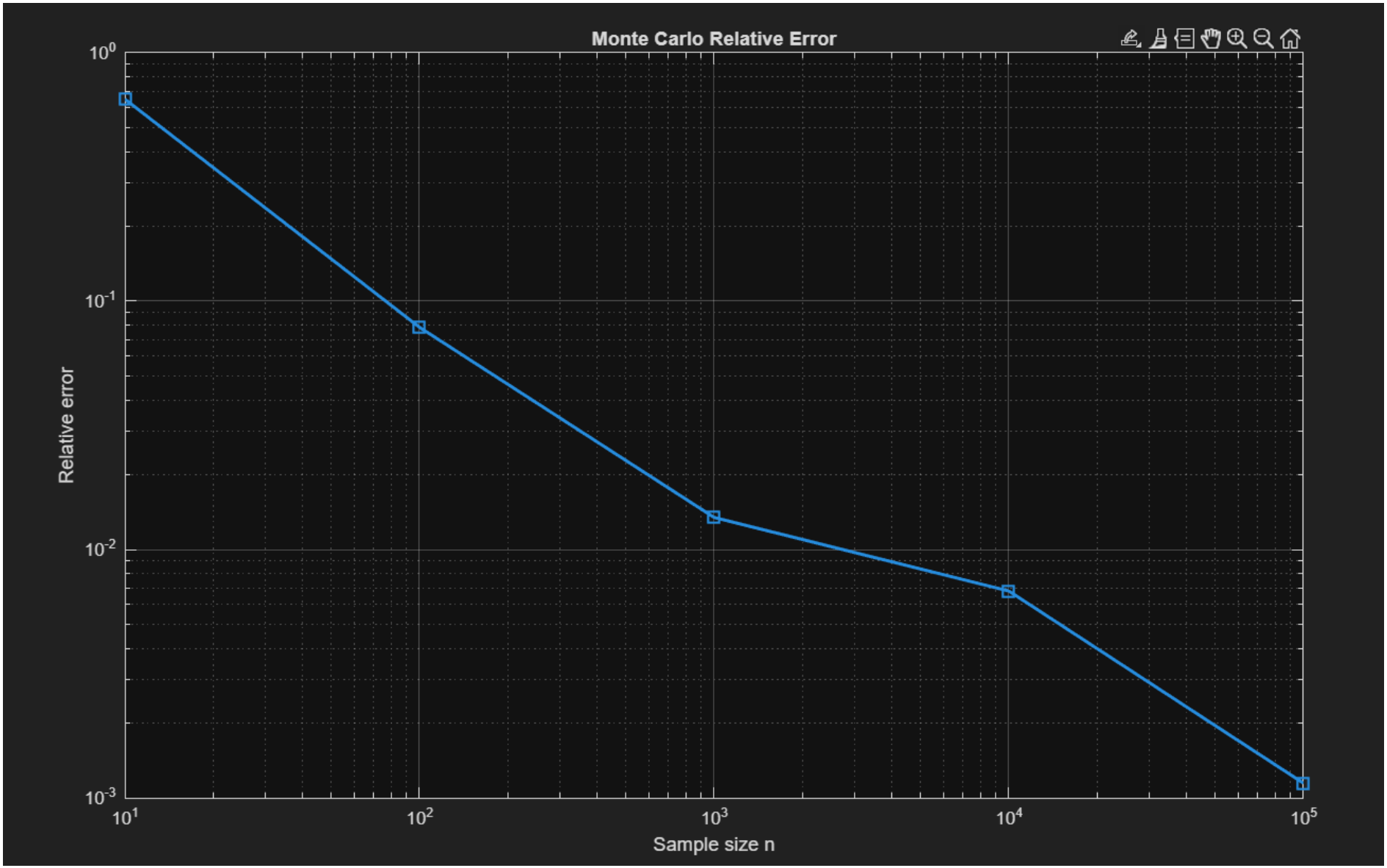
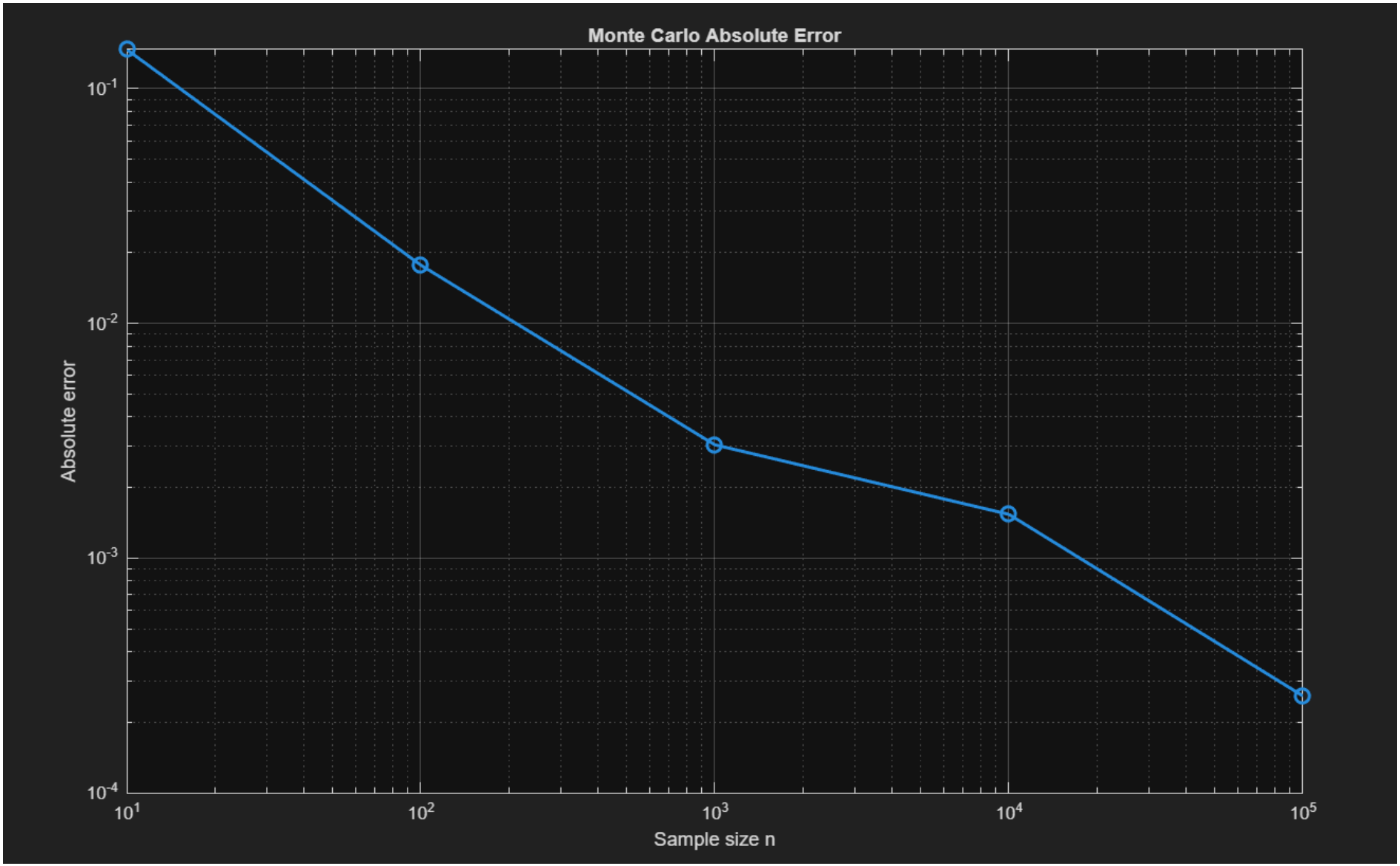
Applying Monte Carlo Methods

Next, we can take a number at random from our defined well-known distributions, substitute that number into our equation and do their for large n and apply the WLLN to get an estimate for the value of our integral.

e.g. Our Uniform [0,1] integral

```
> U = runif(10, min = 0, max = 1); mean( U^3*(sin(U^4)+cos(U^5))*exp(-U^6)
)
[1] 0.1869056
> U = runif(100, min = 0, max = 1); mean( U^3*(sin(U^4)+cos(U^5))*exp(-U^6)
)
[1] 0.2043965
> U = runif(1000, min = 0, max = 1); mean( U^3*(sin(U^4)+cos(U^5))*exp(-U^
6) )
[1] 0.2239863
> U = runif(10000, min = 0, max = 1); mean( U^3*(sin(U^4)+cos(U^5))*exp(-U^
6) )
[1] 0.2240472
> U = runif(100000, min = 0, max = 1); mean( U^3*(sin(U^4)+cos(U^5))*exp(-U
^6) )
[1] 0.2266336
```

MATLAB Error Analysis (Relative)



MATLAB Error Analysis (Absolute)